Shape versus circuit: why graphene geometric diodes and Tesla valves do not make terahertz rectennas

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Geometric diodes, in which an asymmetric channel is meant to rectify by its shape, and electronic Tesla valves have both been proposed as junction-free rectifiers for terahertz rectennas in graphene. We test the proposal in three transport limits with one set of device geometries and validated solvers. A semiclassical ballistic model reproduces the measured asymmetry of a funnel diode in the clean limit and gives none for any Tesla geometry. Steady viscous electron flow makes the classical Tesla valve work, but only at Reynolds numbers above the drift-velocity limit of the Fermi liquid. At terahertz frequencies the gated electron fluid is plasmonic: a funnel matched to a 75 Ω antenna responds at 0.07 to 0.34 A/W, and nearly all of it is the Dyakonov-Shur asymmetry of the source and drain terminations, which any gated channel has. The rectified power follows the square law up to the gate overdrive and peaks below one per cent conversion efficiency for every carrier density and gate distance; lowering either only moves the saturation power, from a milliwatt to a microwatt. Driving the ballistic funnel at terahertz frequencies closes the remaining route: its shape asymmetry survives the transit time only if the potential drop is confined to the neck, and then merely equals the plasmonic figure. The graphene funnel is a picowatt-per-root-hertz square-law detector, not an energy harvester, and the Tesla valve is neither.

I. INTRODUCTION

A rectenna converts radiation to direct current by feeding an antenna into a rectifier. Above a few hundred gigahertz the rectifier is the difficulty: the RC time of a junction diode limits Schottky rectifiers to the low terahertz range, and the search for a junction-free, low-capacitance nonlinearity that works to 1 THz and beyond is long standing. Graphene has attracted two proposals of this kind, both of which rest on the shape of a channel rather than on a junction.

The first is the geometric diode [1–3]: a channel that narrows asymmetrically, typically a funnel, whose dimensions are below the mean free path so that carriers are steered ballistically by the walls. With planar geometry and negligible capacitance it was proposed explicitly for terahertz and infrared rectennas [2]. Encapsulated devices reach reported asymmetry ratios of 1.6 to 2 [4, 5], and related four-terminal ballistic rectifiers have been used for detection and imaging up to 0.7 THz [6, 7]. The second is the electronic Tesla valve. Tesla's valvular conduit [8] rectifies fluid flow by inertia [9], and in the temperature window where electron-electron collisions make the graphene electron liquid viscous [10, 11] a Tesla-shaped channel has been reported to show a five per cent asymmetry of its four-terminal resistance [12]. A shape that rectifies an electron fluid with no junction is an obvious rectenna candidate.

What the shape can do is constrained from the outset. For an elastic two-terminal conductor the unitarity of the scattering matrix implies equal transmission in the two directions at every energy, so the linear response is symmetric whatever the shape. Rectification is a second-order effect, and a consistent theory of it requires the response of the potential landscape inside the device to the bias [13–16]. The useful question is therefore not whether a shape rectifies but which nonlinearity it feeds, how strong that nonlinearity is at the power an antenna delivers, and how fast it is. A geometric diode in the ballistic regime feeds the bias-induced landscape; a Tesla valve in the viscous regime feeds the convective term of the Navier-Stokes equation; and at terahertz frequencies a gated graphene channel is a plasmonic cavity whose convective and density nonlinearities rectify by the mechanism of Dyakonov and Shur [17, 18], which is the basis of most graphene terahertz detectors to date [19, 20]. These three mechanisms have been studied separately, with different geometries and different figures of merit. Whether a geometric shape, as opposed to the terminations of the channel, carries the terahertz response has not been tested with one device in all of them.

This paper does that. We take one set of geometries (Fig. 1): the funnel of the best-documented geometric diode [4], the classical Tesla valve at the same minimum feature size, a straight control, and versions scaled to a 1 μm channel that a 75 Ω antenna can drive. We compute their rectification (i) in a semiclassical ballistic model of the kind used in the geometric-diode literature [2, 21, 22], (ii) in steady viscous electron hydrodynamics, (iii) in the gated compressible (plasmonic) electron fluid from 10 GHz to 3 THz, at small signal and at finite amplitude up to the gate overdrive, and (iv) in the ballistic model driven at terahertz frequencies. Each solver is validated against analytic limits and benchmarks before it is used; the acceptance tests are listed in the Supplemental Material.

The answer is the same in every regime, and it is negative for the shape. The Tesla valve does not rectify ballistically, rectifies viscously only at Reynolds numbers the Fermi liquid cannot sustain, and contributes 10^{-5} to 10^{-3} A/W at terahertz frequencies. The funnel rectifies in all regimes, but its terahertz response of 0.1 to 0.3 A/W is the Dyakonov-Shur asymmetry of its terminations; the shape adds a factor of order one near a taper

resonance and nothing elsewhere. The conversion efficiency of the plasmonic element is proportional to the incident power up to a saturation power set by the gate, and its peak is below one per cent at every operating point for which the fluid description holds. The ballistic shape asymmetry, driven at 1 THz, falls to the same responsivity. As a detector the element is respectable; as a harvester it does not exist.

Section II describes the devices and the three models, Sec. III the dc results, Sec. IV the plasmonic element from small signal to saturation, Sec. V the ballistic element at terahertz frequencies, and Sec. VI compares the regimes, places the numbers against measured detectors and states the limitations.

II. DEVICES AND MODELS

A. Geometries

All devices are two-terminal planar channels with the source contact on the left and the drain on the right. The funnel [Fig. 1(a)] has the dimensions of device D of Nguyen *et al.* [4]: leads 2 μm wide, a taper at 30 degrees to the axis down to a neck of 175 nm, and an abrupt shoulder on the drain side; its measured asymmetry is 1.66 at 80 μA. The classical Tesla valve [Fig. 1(b)] is built from channel centre lines: a zig-zag main channel of width 175 nm inclined at ± 25 degrees, and at every vertex a loop that continues the arriving segment straight on, turns through a half circle of radius 175 nm and rejoins the main channel against the stream. Flow from left to right follows the zig-zag; flow from right to left is split into the four loops. We also keep three earlier variants in which the loops are attached to one side of a straight channel, because they illustrate that the conclusion does not depend on the details of the valve, and a straight 175 nm channel between the same leads as a control. For the terahertz calculations each shape is also scaled so that its narrowest width is 1 μm, with 4 μm contacts, which brings the device resistance from kilohms to the hundred-ohm range of an antenna [Fig. 1(c)].

B. Ballistic model

The ballistic model is the semiclassical trajectory (Monte Carlo [23]) method of the geometric-diode literature [2, 21, 22]. Electrons with the Dirac dispersion $E = \hbar v_F |\boldsymbol{k}|$, $v_F = 10^6$ m/s, are injected at the contacts with the flux-weighted cosine law, fly along classical trajectories, reflect from the walls specularly with probability p and diffusely otherwise, and suffer isotropic momentum-relaxing collisions after exponentially distributed free paths of mean λ. The current is the Landauer expression

$$I = \frac{ge}{h} \int dE\, N(E)\, T(E;V) \left[f_S(E) - f_D(E) \right], \qquad (1)$$

with $g = 4$, $N(E) = EW/\pi\hbar v_F$ the number of modes of the injecting contact of width W, T the transmission probability estimated from the trajectories, and $f_{S,D}$ the Fermi functions of the contacts at electrochemical potentials E_F and $E_F - eV$. Energies are sampled from the bias window $N|f_S - f_D|$, so the estimator has zero variance at zero bias.

With no potential inside the device, flux conservation gives $N_S T_{SD} = N_D T_{DS}$ at every energy and Eq. (1) is odd in V for any shape; the trajectories cannot rectify. Rectification enters through the potential energy $U(\boldsymbol{r}; V)$ that the bias creates inside the channel [13]. We use the simplest assumption of the published simulations, a uniform field along the channel, $U = -eVx/L$ over the device length L, and integrate the curved flights in chords with a controlled sagitta. Self-consistent electrostatics is known to roughly halve the asymmetry of such diodes relative to the uniform-field value and to reverse it for shallow tapers [22]; we return to the field distribution in Sec. V. The figure of merit is the asymmetry $A = I(+V)/|I(-V)|$; writing $I = G_1 V + G_2 V^2$, $A \simeq 1 + 2(G_2/G_1)V$, and a cosine drive of amplitude V_a rectifies $I_{\rm dc} = G_2 V_a^2/2$ if the response is instantaneous. Carriers are electrons at density $n = 1.4 \times 10^{12}\,\mathrm{cm}^{-2}$ ($E_F = 0.138$ eV) and 300 K, the density we estimate from the gate voltage and lever arm reported for that device [4]; holes give the inverse asymmetry.

The solver was validated on eleven tests before use: unit transmission of a specular channel, the free-molecular transmission factors of a diffuse channel [24], the Sharvin conductance [25], the ballistic-to-diffusive crossover, reciprocity with and without the field, unit asymmetry of a mirror-symmetric constriction, and convergence of the chord integration. Classical trajectories require features well above the Fermi wavelength, 30 nm here.

C. Viscous model

Where electron-electron collisions are the fastest scattering process the electron liquid obeys the Navier-Stokes equation [10, 26]. The dc model is steady incompressible flow with momentum relaxation,

$$\mathrm{Re}\,(\boldsymbol{u}\cdot\nabla)\boldsymbol{u} = -\nabla p + \nabla^2 \boldsymbol{u} - \mathrm{Da}\,\boldsymbol{u}, \qquad \nabla\cdot\boldsymbol{u} = 0, \quad (2)$$

in units of the lead width W and the mean inlet speed, with

$$\mathrm{Re} = \frac{I}{ne\nu}, \qquad \mathrm{Da} = \frac{W^2}{\nu\tau_{\rm mr}}, \qquad S = \frac{\ell_s}{W} \qquad (3)$$

the Reynolds number, the damping (Gurzhi) number and the slip length ℓ_s of the Navier wall condition [27] in units of W. In two dimensions the Reynolds number depends on the current and not on the width. The device is current biased with a developed inlet profile, and the figure of merit is the diodicity $D = \Pi_r/\Pi_f$, the ratio of the pressure (voltage) drops for flow from drain to

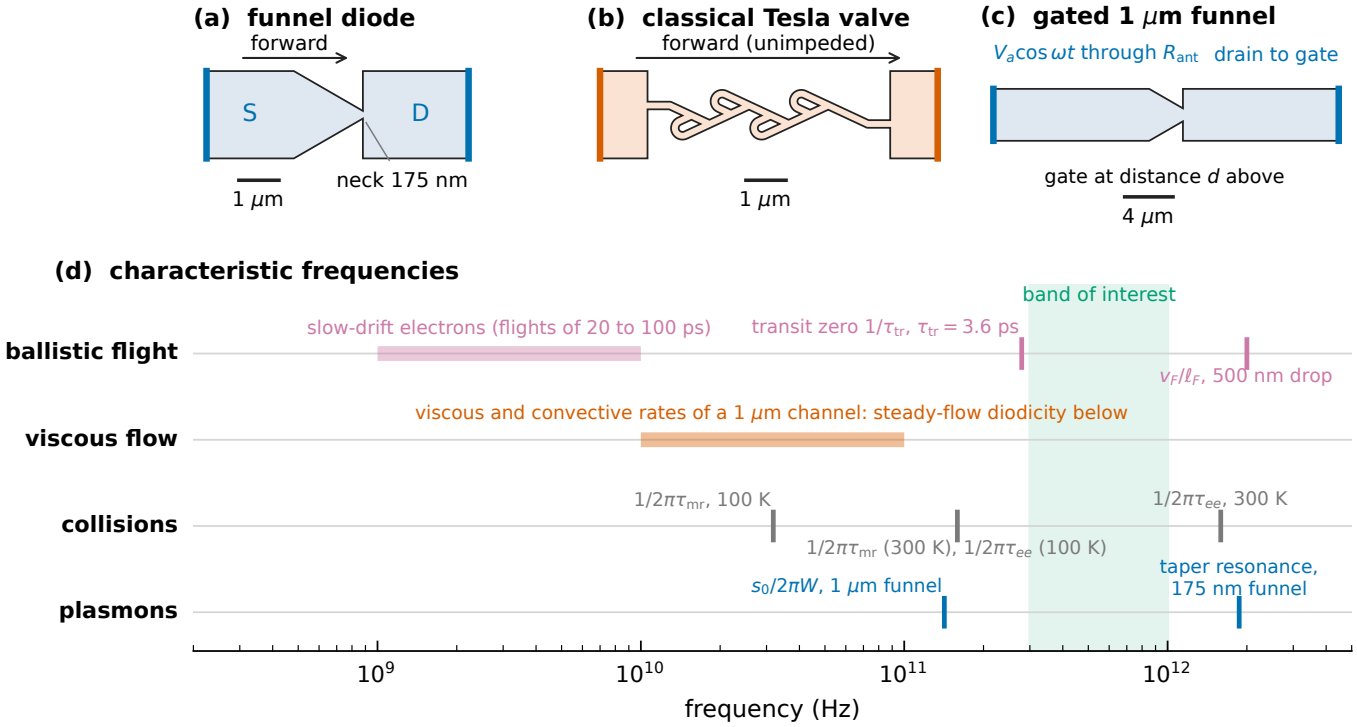


FIG. 1. Devices and frequency scales. (a) The funnel geometric diode, device D of Ref. [4]: 2 μm leads, a 30 degree taper to a 175 nm neck and an abrupt shoulder; forward is the direction into the taper. (b) The classical four-loop Tesla valve at the same 175 nm channel width. (c) The gated funnel scaled to a 1 μm neck with 4 μm contacts, with the terminations used for the plasmonic response: the antenna, a Thevenin source of resistance $R_{\rm ant}$, drives the source contact and the drain is tied to the gate. (d) Characteristic frequencies of the mechanisms studied, for the parameters of Table I; the shaded band is the range of interest for a terahertz rectenna.

source and from source to drain. Equations (2) are solved with Taylor-Hood elements [28] on triangular meshes [29] with Newton iteration and a sparse direct solver [30]; the wall condition $\boldsymbol{u}\cdot\boldsymbol{n}=0$ is imposed with length-weighted normals [31], which makes the discrete wall flux vanish exactly and removes a spurious diodicity of 10^{-5} that polygonal walls otherwise produce. Validation covers Poiseuille and Brinkman flow with slip, Stokes reciprocity ($D-1<10^{-10}$ at $\mathrm{Re}=0$), the Kovasznay solution [32], the backward-facing step [33, 34], and the diodicity of the valve itself as a classical fluid device. The discretisation floor of $D-1$ is 3×10^{-5}.

D. Plasmonic model

At terahertz frequencies the fluid is compressible: a gate at distance d makes the local density follow the local electrochemical potential, and density waves propagate as plasmons. With the gradual-channel approximation and degenerate Dirac carriers the electrochemical potential is $\Phi(n)=e^2n/C_g+\hbar v_F\sqrt{\pi n}$, the hydrodynamic mass is $m^*(n)=\hbar\sqrt{\pi n}/v_F$ [35, 36], and because both depend on the density only the force per unit mass is the gradient of a single function $h(n)$ with $h'=\Phi'/m^*$. The model is

$$\partial_t n+\nabla\cdot(n\boldsymbol{u})=0, \tag{4}$$

$$\partial_t\boldsymbol{u}+(\boldsymbol{u}\cdot\nabla)\boldsymbol{u}+\frac{\boldsymbol{u}}{\tau_{\rm mr}}=-\nabla h(n)+\nu\nabla^2\boldsymbol{u}, \tag{5}$$

with the plasma speed $s(n)$ given by

$$s^2(n)=n\,h'(n)=\frac{e^2n}{C_g m^*(n)}+\frac{v_F^2}{2}. \tag{6}$$

The second term is the quantum capacitance [37]; its share $q=v_F^2/2s_0^2$ of $s_0^2=s^2(n_0)$ is 0.04 at $n_0=10^{12}\,\mathrm{cm}^{-2}$ under 30 nm of hBN and grows to 0.5 at the thin-gate, low-density corner of Sec. IV C. Replacing h by s_0^2n/n_0 gives the shallow-water model of Dyakonov and Shur [17]. At frequencies above $1/\tau_{ee}$ the viscosity is taken in the viscoelastic form $\nu/(1+i\omega\tau_{ee})$.

The antenna is a Thevenin source of amplitude V_a and resistance $R_{\rm ant}$ connected to lumped ports: each contact is an equipotential whose voltage and current satisfy Kirchhoff's law with the external circuit. Two terminations are used throughout. In the *symmetric* one the antenna drives the two contacts in antiphase through $R_{\rm ant}/2$ each and the gate sits at the midpoint; the circuit is then mirror symmetric and any dc response is due to the shape of the channel. In the *Dyakonov-Shur* (DS)

TABLE I. Parameter sets. The ballistic model follows the experiment of Ref. [4]; the two plasmonic sets are representative of hBN-encapsulated graphene at the two temperatures and are labelled by them; they are assumptions, not fits. The 100 K viscosity is of the order measured [10, 11]; the 300 K value is the kinetic estimate $v_F^2\tau_{ee}/4$. W is the contact width.

Quantity	Value
Ballistic	
n, E_F, T	$1.4\times10^{12}\,\mathrm{cm}^{-2}$, 0.138 eV, 300 K
specularity p	1, 0.5, 0
mean free path λ	∞, 700, 200 nm
Viscous	
current at $\mathrm{Re}=1$	160 μA ($\nu=0.1$ m^2/s), 32 μA (0.02)
Da, S	0 to 128; 0 to ∞
Plasmonic	
n_0, d, ϵ_r	$10^{12}\,\mathrm{cm}^{-2}$, 30 nm, 3.8
s_0, gate overdrive	$3.6\,v_F$, 1.43 V
"100 K": τ_{mr}, τ_{ee}, ν	5 ps, 1 ps, 0.1 m^2/s
"300 K": τ_{mr}, τ_{ee}, ν	1 ps, 0.1 ps, 0.03 m^2/s
$W/s_0\tau_{\mathrm{mr}}$ ($W=4\,\mu$m)	0.22 (100 K), 1.12 (300 K)
antenna R_{ant}	75 Ω

termination the antenna drives the source through R_{ant} and the drain is tied to the gate; the terminations are then asymmetric and a uniform channel already rectifies [18]. The responsivity is the short-circuit dc current per available power,

$$R_I=\frac{I_{\mathrm{dc}}}{P_{\mathrm{av}}},\qquad P_{\mathrm{av}}=\frac{V_a^2}{8R_{\mathrm{ant}}},\tag{7}$$

and the Johnson-noise-limited noise-equivalent power is $\mathrm{NEP}=\sqrt{4k_BT/R_{\mathrm{dc}}}/R_I$ with R_{dc} the dc resistance of the device.

Two solvers are used. For small signals the fields are expanded to second order about rest: the first order is a complex time-harmonic plasmon problem, and the time average of the second order is a Stokes-Brinkman problem for the dc flow, forced by products of first-order fields, whose port current or voltage is the rectified signal (Appendix A). For finite amplitudes Eqs. (4) and (5) are integrated in time with the second-order backward difference formula and Newton iteration, with a bias tee that separates the dc load from the antenna. Both use Taylor-Hood elements on the meshes of the viscous model. The validation ladder has seventeen tests: the static limit against the viscous solver, the plasmon dispersion, the Dyakonov-Shur closed form [Fig. 4(a)], reciprocity of the symmetric termination, mesh and time-step convergence, agreement of the two solvers at small amplitude, and invariance under co-scaling of the physical parameters at fixed nondimensional groups.

III. RECTIFICATION AT DC

A. Ballistic: the funnel rectifies, the Tesla valve does not

Figure 2 shows the asymmetry at 0.1 V and the resistance of all six devices for three edge specularities and three mean free paths. In the clean specular limit the funnel gives $A=1.62\pm0.02$ (1.25 at 0.05 V), close to the 1.66 measured at 80 μA [4]; the agreement is conditional, because at the mean free path estimated for these devices [4], 700 nm, the same model gives 1.09, and 1.06 with fully diffuse edges. The uniform-field assumption and the choice of λ and p therefore matter at the factor-of-two level in $A-1$, and we do not claim a quantitative fit. What the comparison establishes is the mechanism and the scale: electrons accelerated toward the neck by the bias arrive with more kinetic energy and more transverse modes than those decelerated toward it.

No Tesla geometry rectifies in this model. All 54 points of the three side-loop variants lie between 0.93 and 1.08 with standard errors of 0.01 to 0.05, and sixteen of the eighteen points of the classical valve are within two standard errors of one. The two exceptions are in the clean specular limit, where the classical valve gives $A=0.93\pm0.02$ at 0.05 V and 0.89 ± 0.01 at 0.1 V: the wrong sign, the blocking direction being the easier one, and gone as soon as edges are partly diffuse or bulk scattering is present. The straight control gives 0.99 ± 0.01. Meanwhile the loops cost resistance in both directions, 1.4 to 22 kΩ against 0.5 to 2 kΩ for the funnel [Fig. 2(b)]. The result follows from the structure of the problem rather than from the parameters: at fixed potential landscape the transmission is reciprocal for any shape, and the Tesla loops do nothing to the way the bias landscape acts on the mode count at a constriction, which is what the funnel exploits.

B. Viscous: the Tesla valve works, beyond the reach of the Fermi liquid

In the electron fluid the picture reverses in principle and not in practice. Figure 3(a) shows the diodicity of the 175 nm classical valve and of the funnel against Reynolds number without damping. The valve is a genuine Tesla valve: with slip walls $D-1$ is 0.15 to 0.23 at $\mathrm{Re}=10$ and 1.1 to 1.5 at $\mathrm{Re}=30$, growing close to Re^2, the signature of an inertial loss present in one direction only. The fields [Fig. 3(c) and 3(d)] show its origin: in the forward direction the loops are stagnant, in the reverse direction each carries a stream that re-enters the main channel against the flow. As a check on the geometry, the same valve treated as a classical no-slip fluid device gives $D=1.05$ at $\mathrm{Re}=50$ and 1.23 at $\mathrm{Re}=100$; three-dimensional simulations of multistage valves give 1.02 to 1.03 and 1.10 to 1.15 at the same Reynolds numbers [38], a duct of finite depth adding wall friction that the two-dimensional

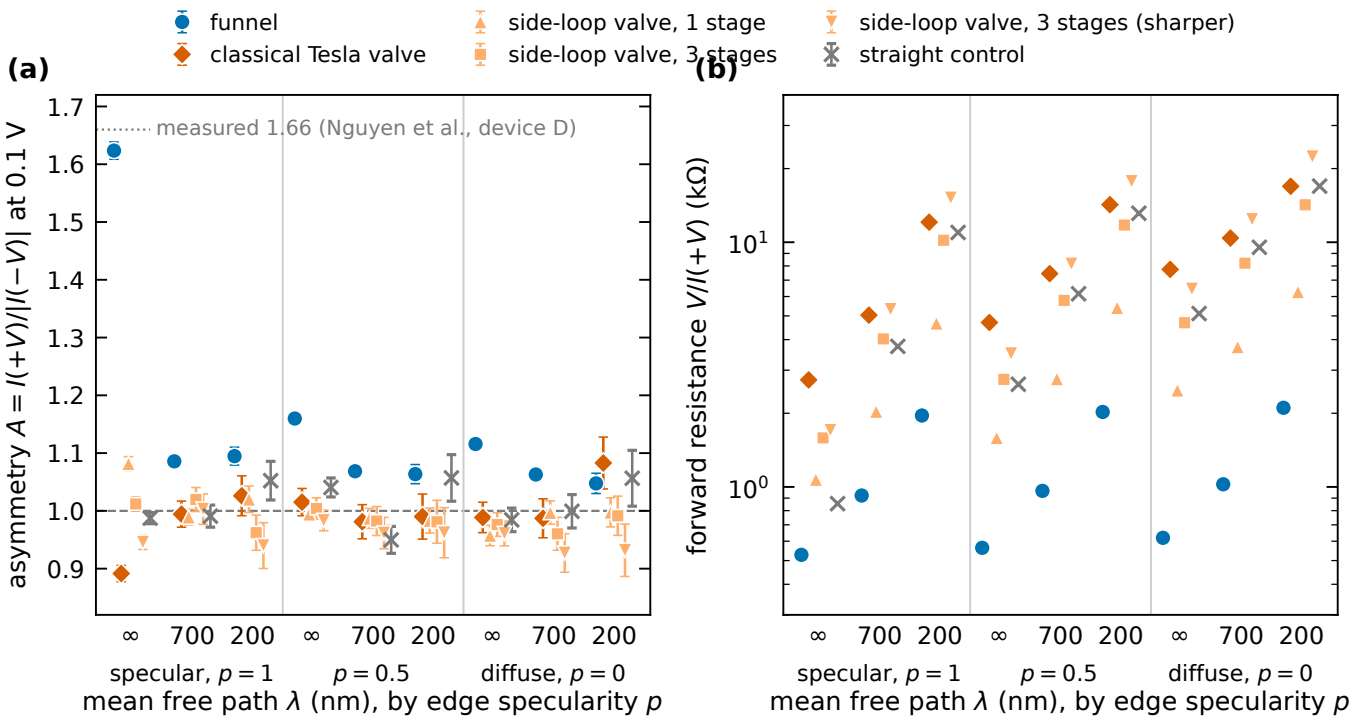


FIG. 2. Ballistic transport at dc. (a) Asymmetry $A = I(+V)/|I(-V)|$ at 0.1 V for the funnel, four Tesla-valve geometries and the straight control, for edge specularities p and mean free paths λ; error bars are one standard error from 64 batches of 400 000 trajectories in total. The dotted line marks the measured value for the funnel [4]. (b) Forward resistance of the same devices.

flow lacks. The diodicity of the valve is converged with respect to the mesh to about 10% (Supplemental Material), which is immaterial for what follows. The funnel is a diode of the opposite sign, the forward direction being the harder one: its asymmetry is that of the exits, a jet from the abrupt shoulder against a diffuser along the taper, and it is the stronger of the two below $\mathrm{Re} \approx 3$ ($|D-1| = 1.2 \times 10^{-2}$ at $\mathrm{Re} = 3$ with slip) but saturates near 0.15.

Three facts decide the matter. First, the experiment of Ref. [12] used currents of at most a few microamperes (6.3 μA for the device that showed the effect), for which we estimate $\mathrm{Re} \approx 0.04$ to 0.2 at $10^{12}\,\mathrm{cm}^{-2}$ and $\nu = 0.1$ to 0.02 m^2/s; the authors quote no Reynolds number. For Re up to 0.3 every geometry, wall condition and damping gives $|D-1| < 3\times10^{-5}$, three orders of magnitude below the reported five per cent, which was observed for holes in a single-loop valve as a ratio of four-terminal resistances. Steady incompressible electron hydrodynamics does not reproduce that observation; whatever produces it is not the inertial Tesla mechanism, and we do not explain it. The candidates that Eq. (2) excludes are the compressibility of the gated fluid and electron heating. Second, the Reynolds number is bounded. The drift speed in a channel of width w is $u = \mathrm{Re}\,\nu/w$, and keeping it below $0.3\,v_F$, where a Fermi-liquid description is still tenable, caps Re at 0.5 to 2.6 in a 175 nm channel and at 3 to 15 in a 1 μm channel for $\nu = 0.1$ to 0.02 m^2/s [shaded in Fig. 3]. Third, the low viscosity needed for a large Reynolds number comes with weak slip relative to the width and strong relative damping. Figure 3(b) shows the valve scaled to 1 μm: each factor of four in Da costs a factor two to three, and a slip length of 0.4 μm instead of perfect slip a factor twenty. Inside the admissible window the diodicity is 10^{-3} to 3×10^{-2}, at a resistance of 0.3 to 1.6 kΩ; the 175 nm valve costs 10 to 86 kΩ for less.

The steady-flow diodicity is also slow. It is established over the viscous and convective times of the channel, 10 to 20 ps at 1 μm, so the picture of this subsection applies below some tens of gigahertz. Above that the gate makes the fluid compressible, and the question has to be asked of a different model.

IV. THE PLASMONIC ELEMENT AT TERAHERTZ FREQUENCIES

A. Small signal: the terminations rectify, the shape tunes

The solver is first checked where an analytic answer exists. For a uniform channel driven by an ideal source with an open drain, the shallow-water closure reproduces the closed form of Dyakonov and Shur [18] to three digits from 10 GHz to 3 THz, through the plasmon resonances at 100 K and on the broadband plateau at 300 K [Fig. 4(a)]. The Dirac closure of Eq. (6) raises the response by 40% near the resonances and by up to a factor

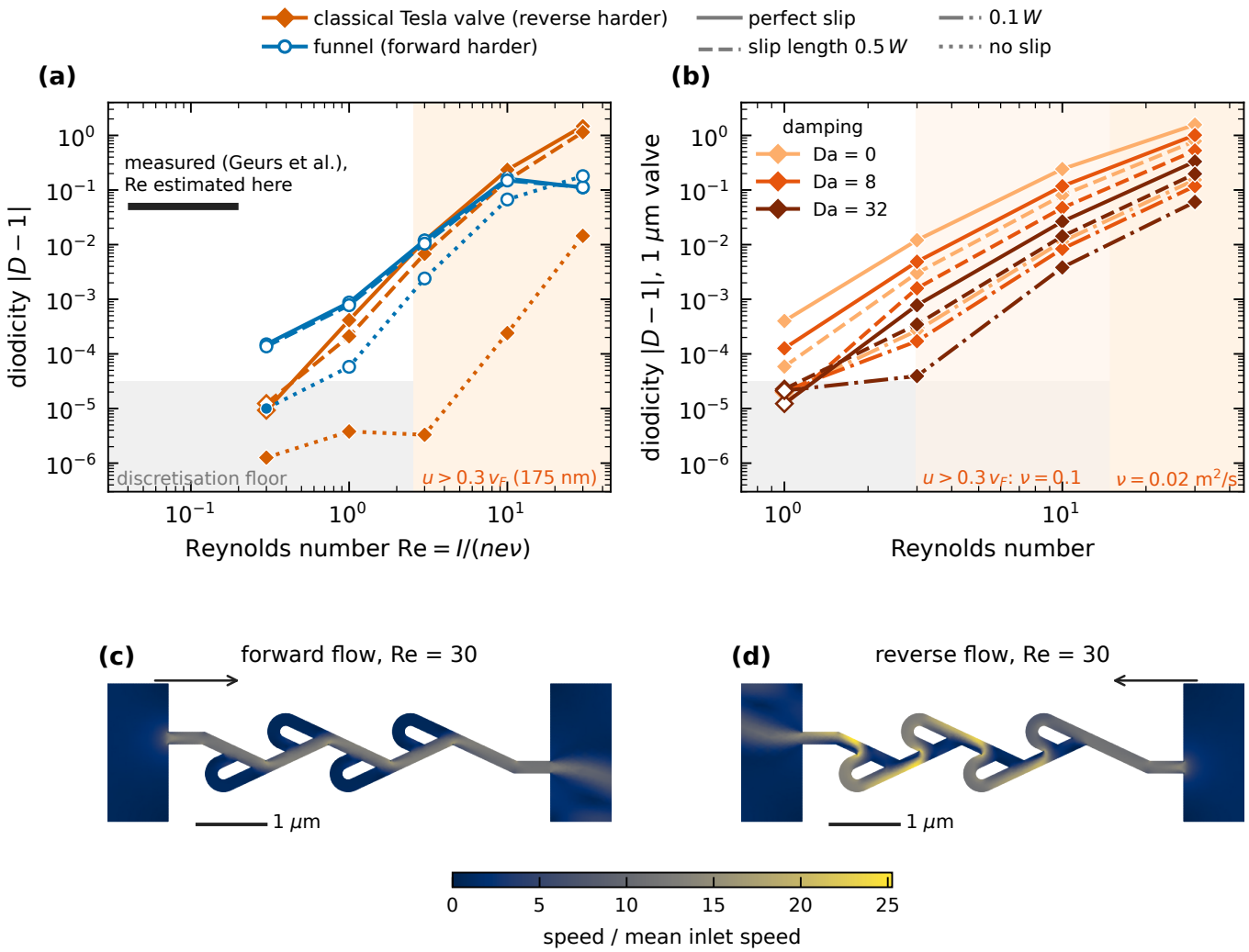


FIG. 3. Steady viscous flow. (a) Diodicity $|D-1|$ of the 175 nm classical Tesla valve and of the funnel without damping, for three wall conditions; filled symbols: reverse direction harder ($D>1$), open symbols: forward harder. The bar marks the diodicity reported in Ref. [12] at our estimate of its Reynolds number; grey: below the discretisation floor; orange: drift speed above $0.3\,v_F$. (b) The valve scaled to a 1 μm channel, with momentum-relaxation damping Da and finite slip length; the two shaded ranges are the drift-speed limits for $\nu = 0.1$ and 0.02 m^2/s. (c), (d) Speed in the 175 nm valve at Re = 30 with perfect slip for forward and reverse flow.

two between them; the difference is the nonlinearity of the Dirac fluid itself, whose density-dependent mass and quantum capacitance add second-order terms that the shallow-water model lacks.

Figure 4(b) shows the 1 μm funnel. Its impedance is 80 to 200 Ω across the band, so it absorbs 80 to 95% of the available power of a 75 Ω source at 100 K. With the DS termination the responsivity is 0.15 to 0.34 A/W at 100 K and 0.07 to 0.09 A/W at 300 K, flat from 10 GHz to beyond 1 THz. With the symmetric termination, which isolates the shape, it is 10^{-5} to 10^{-2} A/W below 100 GHz, rises to 0.13 A/W at 1 THz at 100 K as the taper becomes comparable to the plasmon wavelength, and stays at a few mA/W at 300 K, where friction and viscosity damp the plasmon. Below 100 GHz the shape-only response is the curvature of the steady current-voltage relation of Sec. III; a ladder test ties the two solvers together there. The sign of the DS response follows the termination: exchanging the roles of source and drain reverses it.

The 175 nm funnel [Fig. 4(c)] has a dc resistance of 1.9 kΩ and absorbs only 15 to 80% of the available power, but it has one feature the wide device lacks. At 1.87 THz the taper passes through a series resonance, the horn is matched to the antenna, and the shape-only responsivity reaches 0.22 A/W at 100 K, half of the 0.45 A/W of the DS termination at the same frequency. The resonance belongs to the taper and not to the leads: a control with 1 μm leads has it at the same frequency, with an impedance of $57-8i$ Ω there. This is the one place where the funnel shape earns its keep in a plasmonic design, as a matched horn. At 300 K it gives 0.03 A/W against 0.10 A/W for the terminations.

The Tesla valve [Fig. 4(d)] contributes almost nothing. Its shape-only responsivity is 10^{-5} to 10^{-3} A/W below 300 GHz and reaches 2×10^{-2} A/W only at a 100 K resonance near 0.7 THz, ten to twenty times below the funnel at every frequency and temperature. Its resistance of 6 to 11 kΩ is two orders of magnitude from the antenna, so 3 to 50% of the available power is absorbed, and whatever the DS termination adds, 0.02 to 0.1 A/W, is the terminations and not the loops. The valve's asymmetry is an inertial loss that needs a Reynolds number of ten; at terahertz frequencies the long path that produces it is also several plasmon wavelengths long, so its stages do not act coherently.

With ideal contacts and the Johnson noise of $R_{\rm dc}$ alone, the noise-equivalent power of the funnel elements is 4 to 60 pW/Hz$^{1/2}$ between 0.3 and 2 THz: 17 pW/Hz$^{1/2}$ for the 1 μm funnel at 100 K, about 60 pW/Hz$^{1/2}$ at 300 K, and 4 pW/Hz$^{1/2}$ for the 175 nm funnel at its taper resonance at 100 K.

B. Large signal: a square law up to the gate overdrive

A detector is judged by responsivity and noise, a harvester by conversion efficiency $\eta = P_{\rm dc}/P_{\rm av}$ at the power the antenna supplies. In second-order perturbation the device is a dc Thevenin source with open-circuit voltage $V_{\rm oc} = r\,V_a^2/V_u$ and resistance $R_{\rm dc}$, where $V_u = m^* s_0^2/e$ is the natural voltage unit of the gated fluid (1.49 V here, the gate overdrive up to the quantum-capacitance correction) and r is a pure number. Into a matched dc load

$$\eta = 2r^2\,\frac{R_{\rm ant}}{R_{\rm dc}}\,\frac{V_a^2}{V_u^2} = \frac{16\,r^2 R_{\rm ant}^2}{R_{\rm dc}V_u^2}\,P_{\rm av}. \qquad (8)$$

For the 1 μm funnel $r\approx0.1$ and $\eta/P_{\rm av}$ is 1.0 to 2.4 W^{-1}. Since V_a cannot usefully exceed V_u, where the channel is depleted on the negative half cycle, Eq. (8) already bounds the efficiency below one per cent; what the nonlinear solver decides is where the law fails and how.

Figure 5 shows the time-domain results for the DS-terminated 1 μm funnel. The efficiency meets the perturbative law to 0.3% at the smallest amplitude and stays within about 10% of it up to 0.9 mW of available power at 0.1 and 0.3 THz at both temperatures. At 1 THz the response saturates first, 10% below the law at 150 μW at 100 K. At 1.8 mW, a swing of 1.04 V on a gate overdrive of 1.43 V, the efficiency is 0.14 to 0.38% and the response is 0.6 to 0.9 of the square law at 100 K; at 300 K the overdamped device is weakly super-quadratic, 1.10 to 1.14 of the law at 0.1 and 0.3 THz. The density inside the device swings between 0.52 and 1.6 of its gated value [Fig. 5(c)] and the solver converges at every step, so the saturation is a property of the fluid and not a failure of the method. Returning the dc current through the antenna instead of a matched load costs the expected 15% and nothing more.

The shape-only response behaves worse, not better, at large signal. With the symmetric drive the funnel gives 0.56 W^{-1} at 0.3 THz, a third of the DS term of the same device; it leaves its square law at 150 μW and is at 0.43 of it at 0.9 mW [Fig. 5(b)]. The picture of a geometric diode that improves when driven hard is not what the fluid does: the constriction's rectification is a resonance effect that detunes as the density swings.

The scale of the result is set by Eq. (8). A beam of 1 W/m^2 delivers about 10 nW to a half-wave antenna at 1 THz, which the element converts at $\eta\sim10^{-8}$; a full milliwatt focused on one element is converted at 0.2%.

C. Operating point and contacts: the knee moves, the ceiling does not

Equation (8) suggests a lever. The efficiency saturates when V_a approaches V_u, at an available power $P_{\rm knee} = (V_u/2)^2/8R_{\rm ant}$, and V_u is proportional to the gate overdrive $en_0/C_g \propto n_0 d$. A lower density or a thinner gate should bring the saturation down to the microwatts an antenna collects. What it does to the peak depends on the pure number $r^2R_{\rm ant}/R_{\rm dc}$, which the solver has to supply. We swept $n_0 = 5\times10^{10}$ to 10^{12} cm^{-2} and $d = 30$, 10, 5 nm with the scattering times of Table I, flagging points outside the Fermi-liquid limit $E_F \geq 3k_BT$ or the gating limit $en_0/C_g \geq 4k_BT/e$; the saturation factors of Sec. IV B were carried over in units of V_u (Appendix B), and a second time-domain ladder at $(5\times10^{11}\,{\rm cm}^{-2}, 10\,{\rm nm})$ confirmed that they transfer, to 0.2% at 300 K and within 14% at 100 K.

The saturation power moves as predicted [Fig. 6(a)]: from 0.92 mW at the reference point to 33 μW at $(5\times10^{11}\,{\rm cm}^{-2}, 10\,{\rm nm})$ and 3 μW at $(5\times10^{10}\,{\rm cm}^{-2}, 30\,{\rm nm})$, with $\eta/P_{\rm av}$ rising from 1.9 to 51 and 1770 W^{-1} at 100 K and 0.3 THz. The peak barely moves [Fig. 6(b)]. Over the valid grid it is 0.03 to 0.9% at 100 K and at most 0.24% at 300 K, where the reference point is already the best. A thinner gate lowers it, because r falls from 0.088 to 0.050 between 30 and 5 nm as the plasma speed drops and friction and quantum capacitance gain weight; a lower density raises r, to 0.32 at the edge of the Fermi-liquid limit. A matched antenna in place of 75 Ω gives at best 1.3% at 100 K and 0.27% at 300 K. If the mobility rather than the scattering time is held constant as the density falls, the low-density gain shrinks by a factor four to six. The design rule is therefore to put the knee at the available power; the element then converts at a few tenths of a per cent, and nothing in n_0 or d raises that number.

Contact resistance is the dominant practical loss, and it is not a series loss. A contact resistance R_c at each contact of the reference device reduces $\eta/P_{\rm av}$ by factors 1.6, 4.6 and 78 for $R_c = 100$, 300 and 1000 Ω μm at 100 K and 0.3 THz [Fig. 6(c)], the responsivity from 0.21 A/W to 0.15, 0.07 and 0.01 A/W, and it raises the NEP from 27 to 35, 59 and 240 pW/Hz$^{1/2}$. The series picture un-

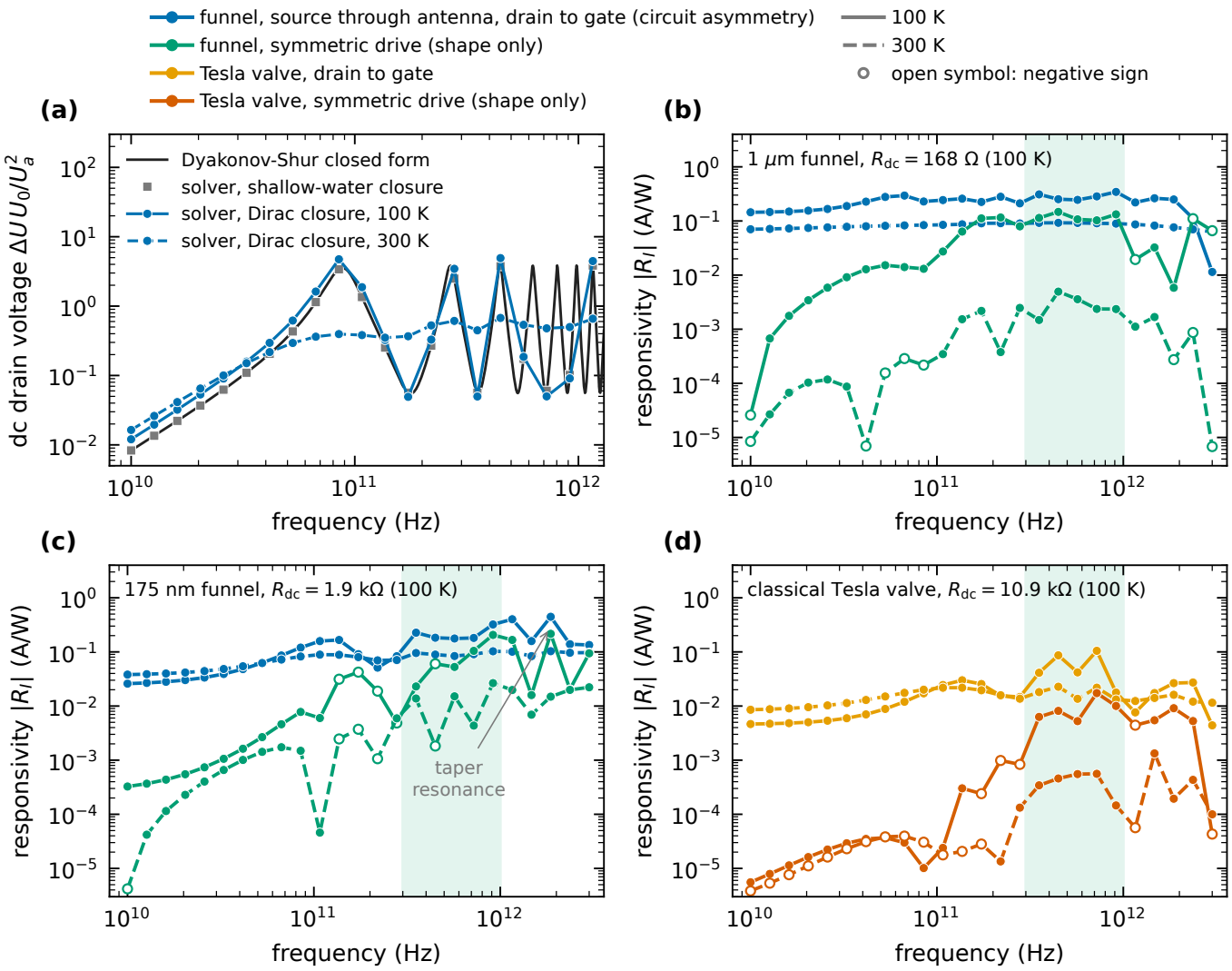


FIG. 4. Small-signal plasmonic response. (a) Validation on a uniform 1 μm × 10 μm channel with an ideal source and open drain: dc drain voltage per squared ac amplitude in units of the inverse gate overdrive. The solver with the shallow-water closure lies on the Dyakonov-Shur closed form [18]; the Dirac closure of Eq. (6) is 40% higher near resonance. (b)–(d) Current responsivity into a 75 Ω antenna for the 1 μm funnel, the 175 nm funnel and the classical Tesla valve, with the drain tied to the gate (circuit asymmetry) and with the symmetric drive that isolates the shape. Open symbols mark a negative sign. The shaded band is 0.3 to 1 THz.

derestimates this. The gate carries displacement current, so the ac drain current is not the negative of the source current, and a resistance at the drain lifts it off the gate potential at the signal frequency: the boundary condition that makes the Dyakonov-Shur mechanism work is partly removed. Matching the antenna to the new impedance recovers little.

For detection the same lever is worth having [Fig. 6(d)]. The Johnson-limited NEP of the 1 μm funnel at 0.3 THz falls from 27 pW/Hz$^{1/2}$ at the reference point to about 1 pW/Hz$^{1/2}$ at 5×10^{10} cm^{-2} at 100 K if the 5 ps scattering time survives there, 2 pW/Hz$^{1/2}$ with constant mobility or with 300 Ω μm contacts; at 300 K the best valid point is 15 pW/Hz$^{1/2}$.

V. THE BALLISTIC ELEMENT AT TERAHERTZ FREQUENCIES

One mechanism is outside the fluid model by construction. In the clean limit the ballistic funnel of Sec. III has a curvature $G_2 = 3.8$ mA/V^2; if it responded instantaneously, a cosine drive of 0.1 V across the device would rectify 18 μA, which is 0.87 A/W of the available power of a 75 Ω source, two to ten times the plasmonic figures. Whether it does respond is a transit-time question, and we ask it of the same model.

The imposed potential is made time dependent, $U(x,t) = -eV_a\cos(\omega t)\,g(x)$, and the flights integrate $\hbar\dot{\boldsymbol{k}} = -\nabla U$ without the energy conservation that holds

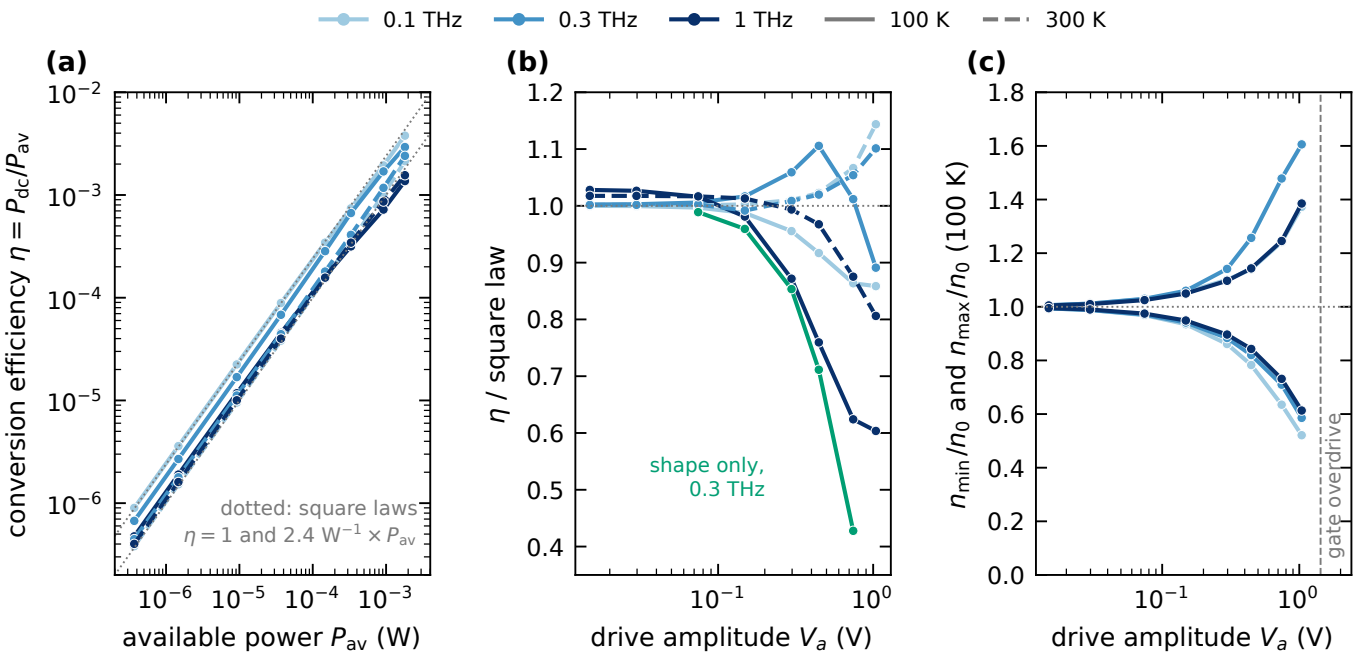


FIG. 5. Finite-amplitude response of the DS-terminated 1 μm funnel from the time-domain solver. (a) Conversion efficiency into a matched dc load against available power. (b) Efficiency relative to the square law of Eq. (8), against the ac amplitude; the green curve is the shape-only response (symmetric drive) at 0.3 THz, 100 K. (c) Smallest and largest density in the device over a period, relative to the gated density, at 100 K.

only in a static landscape. Two profiles g bracket the electrostatics, which the model does not compute: the uniform ramp over the 6 μm device used at dc, and a drop of length ℓ_F centred on the neck, with ℓ_F = 175, 500 and 1600 nm (one neck width, three, and the taper length). The bias-window estimator of Eq. (1) relies on flux conservation in a fixed landscape and fails in a moving one, so electrons are injected from both contacts with the equilibrium distribution $N(E)f(E-E_F)$ at uniformly distributed phases, and

$$I_{\rm dc} = \frac{ge}{h}\,\Phi\,\big[\langle T_{SD}\rangle - \langle T_{DS}\rangle\big], \tag{9}$$

with $\Phi = \int dE\, N(E)f(E-E_F)$ and the averages taken over injection phase; $(ge/h)\Phi$ is 1.43 mA for the 2 μm contacts. The first harmonic of the drain current, from the arrival phases, gives the transport admittance and hence the mismatch to the antenna. Six further tests validate the estimator: it reproduces the bias-window current in a static field, it gives zero at zero bias, zero for the straight control at every frequency, and at 300 MHz it reproduces the rectified current and the admittance computed from the static curve (Supplemental Material). The results are reported as the transfer $S_B(f) = I_{\rm dc}(f)/I_{\rm dc}(0)$, with $I_{\rm dc}(0)$ the quasi-static value for the same field profile, and as responsivity.

A. Uniform field: two roll-offs

With the uniform field the response dies in two steps [Fig. 7(a)]. The transfer is 0.98 ± 0.04 at 0.3 GHz and already 0.82 at 3 GHz and 0.67 at 10 GHz, although a direct flight from the source contact to the neck takes 3.6 ps. The reason is in the distribution of flight times [Fig. 7(c)]. The taper scatters part of the incoming electrons into directions nearly perpendicular to the axis, and these drift back to the source contact, bouncing between the walls of the lead, over tens of picoseconds: 5% of the source-injected electrons fly longer than 20 ps and 0.9% longer than 50 ps. In a field that fills the leads, this slow population is displaced in energy between wall collisions once the period approaches its lifetime, and the transmission in both directions changes. The response time of a ballistic device is set by its slowest population, not by its mean transit, and a converging taper manufactures slow electrons. The second step is the transit roll-off of the direct electrons: the transfer is 0.41 at 100 GHz, crosses zero at 300 GHz, where $\sin(\pi f \tau_{\rm tr})$ vanishes for $\tau_{\rm tr}$ = 3.6 ps, and is 0.05 ± 0.04 at 1 THz, i.e. 0.05 A/W. At a mean free path of 700 nm the quasi-static responsivity is only 0.15 A/W to begin with; it is 0.04 A/W at 30 GHz and zero at 1 THz. With the field model in which the dc asymmetry was established, the ballistic funnel is not a terahertz rectifier.

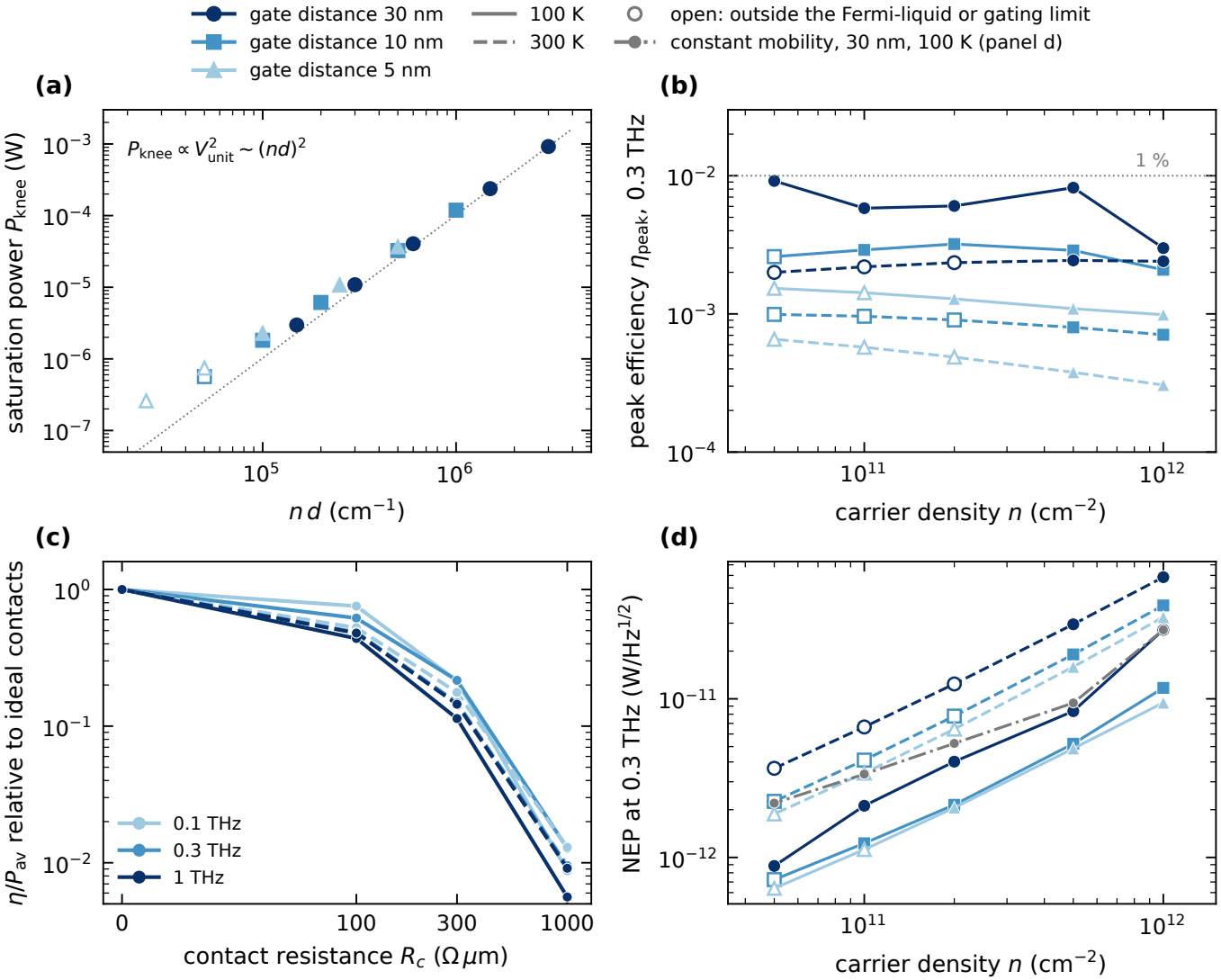


FIG. 6. Carrier density, gate distance and contacts for the DS-terminated 1 μm funnel. (a) Saturation power P_{knee} against the product $n_0 d$; the dotted line is $(n_0 d)^2$ through the reference point, and the departure at small $n_0 d$ is the quantum capacitance. (b) Predicted peak efficiency at 0.3 THz. (c) Efficiency per watt against contact resistance, relative to ideal contacts, at the reference point. (d) Johnson-limited NEP at 0.3 THz. Open symbols are outside the Fermi-liquid or gating limits.

B. Field confined to the neck

Self-consistent electrostatics concentrates the field of such a diode at the neck rather than spreading it along the leads [22]. Confining the drop to the neck lowers the static asymmetry, from 1.61 for the uniform field to 1.52, 1.50 and 1.40 for ℓ_F = 1600, 500 and 175 nm, and it changes the dynamics [Fig. 7(a) and 7(b)]. For ℓ_F = 500 nm the transfer is 1.02 ± 0.04 up to 3 GHz, since the slow electrons in the lead no longer see a field, then falls to 0.54 at 100 GHz, and is 0.64 at 300 GHz and 0.38 ± 0.04 at 1 THz. The decline between 10 and 100 GHz is not what the 0.5 ps crossing of the field region would suggest; the transfer is set by the whole flight between the contacts. At 1 THz the transfer is 0.51 for a drop of one neck width, 0.38 for 500 nm and zero for 1600 nm. The corresponding responsivities are 0.45±0.04 and 0.37 ± 0.04 A/W: this is the most the shape mechanism gives at 1 THz in this model, with infinite mean free path, specular walls and a drop of favourable length, and it equals the Dyakonov-Shur response of the plasmonic funnel at 100 K rather than exceeding it. The amplitude dependence is that of the dc curve: at 300 GHz the rectified current follows V_a^2 to 0.1 V and is 0.40 and 0.24 of the square law at 0.2 and 0.3 V.

Two further cases show that the terahertz response of the confined field is not simply the static asymmetry carried up in frequency. At λ = 700 nm the static asymmetry of the confined-field models is reversed (A = 0.94 to 0.97), the rectified current passes through zero between

30 and 100 GHz, and it returns with the sign of the clean funnel, 0.20 A/W at 300 GHz and 0.24 A/W at 1 THz. With half-diffuse walls ($p = 0.5$) in the clean limit the static asymmetry of the 500 nm drop is 1.01, and the responsivity at 1 THz is nevertheless 0.45 ± 0.04 A/W (0.31 A/W for the neck-width drop). There is, then, a transit-time rectification of the funnel with a neck-confined field that is robust against edge roughness and bulk scattering, of 0.2 to 0.45 A/W. It does not depend on the treatment of the few electrons that reach the Dirac point: replacing their removal by a classical turning point changes no result by more than one standard error.

VI. DISCUSSION

A. One answer in three regimes

Table II and Fig. 8 collect the results. The Tesla valve fails in every regime, and for one reason. Its asymmetry is an inertial loss: it needs a fluid, and a Reynolds number of order ten. The ballistic electron gas is not a fluid, so the loops only add resistance. The viscous electron liquid is one, but its Reynolds number is capped by the Fermi velocity at values where the valve gives a per cent or less. At terahertz frequencies the valve is long compared with the plasmon wavelength and its resistance is two orders of magnitude from an antenna.

The funnel rectifies in every regime, by a different mechanism in each, and the shape is essential in none of the fast ones. At dc in the ballistic regime the shape and the bias landscape together give an asymmetry of up to 1.6; that response is limited to a few gigahertz when the field fills the leads. In the viscous regime the funnel is a weak exit-loss diode. At terahertz frequencies the gated funnel's rectification is that of its terminations: tying the drain to the gate and driving the source produces 0.1 to 0.3 A/W in a funnel, as it would in a plain channel of the same impedance, while the symmetric drive that leaves only the shape produces a comparable response only near a taper resonance at low temperature, and that response saturates first at large signal. The ballistic shape mechanism with a neck-confined field reaches the same 0.2 to 0.45 A/W at 1 THz in its most favourable idealisation. Neither transport limit gives a shape-based rectifier that outperforms the circuit asymmetry of a gated channel.

B. What the element is good for

Table III and Fig. 8(b) place the simulated element against measured graphene terahertz detectors. At room temperature the best reported devices have noise-equivalent powers of 100 to 130 pW/Hz$^{1/2}$ referred to the optical power at 0.4 THz [39, 40] and 80 to 160 pW/Hz$^{1/2}$ referred to a diffraction-limited spot at 2.5 to 2.8 THz [41, 42]; most others lie between 0.5 and 2 nW/Hz$^{1/2}$ [43–46], the first devices were at tens of nW/Hz$^{1/2}$ [19], and a four-terminal ballistic rectifier has been reported at 34 pW/Hz$^{1/2}$ [7]. The Johnson-limited 60 pW/Hz$^{1/2}$ of the DS-terminated 1 μm funnel at 300 K, or 150 pW/Hz$^{1/2}$ with 300 $\Omega\,\mu$m contacts, is therefore where the best measured devices already are, and its voltage responsivity of 50 V/W is within their range of 1 to 100 V/W. The idealised model holds no hidden order of magnitude; the comparison supports the model rather than promising an improvement. At low temperature, encapsulated devices reach 0.2 pW/Hz$^{1/2}$ at 10 K and 0.13 THz [20, 47] and 18 pW/Hz$^{1/2}$ at 77 K and 2.8 THz [42]; the simulated 17 to 27 pW/Hz$^{1/2}$ at 100 K, and 1 pW/Hz$^{1/2}$ at low density, lie in the same range. Three caveats apply. Nearly all reported values, like ours, are computed from the Johnson noise of the measured resistance and not from measured noise. The reference power differs between reports (total beam, power delivered to the antenna, or power scaled to a diffraction-limited spot), and values normalised to the active device area are not comparable and are left out. And part of the measured response of graphene detectors is photothermoelectric [40, 41, 48], which the model does not contain. The room-temperature incumbent is better than any of these: commercial zero-bias Schottky detectors have typical noise-equivalent powers of 5 to 27 pW/Hz$^{1/2}$ from 0.22 to 1.1 THz [49], which the simulated graphene funnel reaches only at 100 K.

For energy conversion the conclusion does not depend on such comparisons. The efficiency of the plasmonic element is bounded by the pure number $r^2 R_{\mathrm{ant}}/R_{\mathrm{dc}}$ of Eq. (8) at an amplitude equal to the gate overdrive. That number is of order 10^{-2} and no density or gate distance in the hydrodynamic window raises it; a low density or a thin gate moves the power at which the bound is reached from a milliwatt to a microwatt, which is useful for matching a detector's dynamic range to its signal, and nothing more. Realistic contacts of 100 to 300 $\Omega\,\mu$m cost a further factor two to five. A rectenna made of such elements has no harvesting regime.

A designer who wants a graphene terahertz detector from these shapes should therefore read the results as follows. Use the funnel as a matched horn, not as a diode: its value is the impedance it presents and the taper resonance. Terminate the channel asymmetrically, since that is where the rectification comes from. Put the saturation knee at the expected power by choice of density and gate distance. Spend the fabrication effort on the contacts, because a resistive drain contact removes the asymmetry it is supposed to serve. Do not add Tesla loops.

C. Limitations

Every idealisation in this work favours the device, which is the right direction for a negative result but should be stated. There is no experiment here: the solvers are validated against analytic limits and one measured number, and the agreement with that number

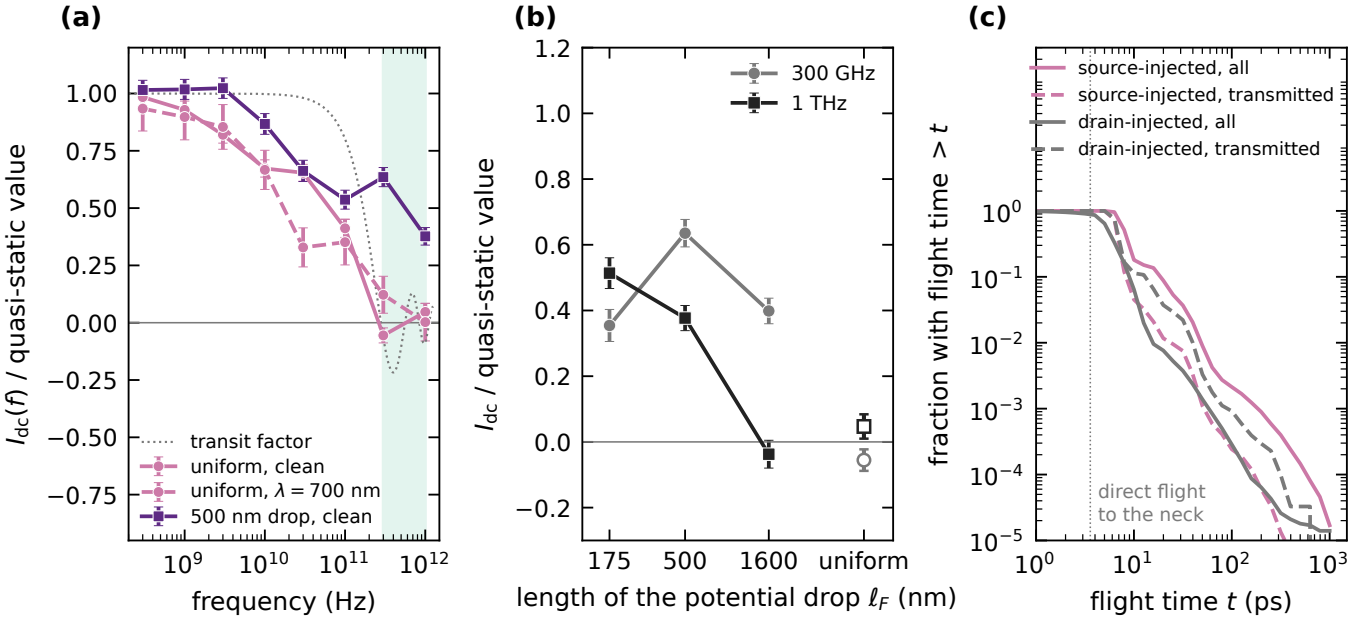


FIG. 7. The ballistic funnel under an ac drive of amplitude 0.1 V. (a) Rectified current relative to its quasi-static value for the uniform field (clean limit and $\lambda = 700\,\mathrm{nm}$) and for a potential drop of 500 nm at the neck; the dotted line is the transit factor for a flight of 3.6 ps from the source contact to the neck. (b) The same ratio at 0.3 and 1 THz against the length of the drop; open symbols: uniform field. (c) Fraction of electrons with a flight time longer than t in the static device at 0.1 V. Error bars are one standard error.

TABLE II. Rectification by geometry and regime. A: ballistic asymmetry at 0.1 V; D: viscous diodicity; R_I: current responsivity into a 75 Ω source. "Shape only" is the symmetric termination; DS has the drain tied to the gate.

Regime	Funnel	Classical Tesla valve	What limits it
Ballistic, dc	$A = 1.62$ clean; 1.06 to 1.09 at $\lambda = 700\,\mathrm{nm}$	$A = 0.89$ to 1.08, no systematic sign	reciprocity; needs the bias landscape
Viscous, dc	$\lvert D-1\rvert \le 10^{-2}$ within the drift limit	10^{-3} to 3×10^{-2} within the drift limit	drift speed below $0.3\,v_F$ caps Re
Plasmonic, shape only	10^{-4} to 0.13 A/W; 0.22 A/W at taper resonance	10^{-5} to 10^{-3} A/W; 0.02 A/W at resonance	plasmon damping; saturates first
Plasmonic, DS	0.07 to 0.34 A/W; $\eta < 1\%$	0.02 to 0.1 A/W at 6 to 11 kΩ	r^2; gate overdrive; contacts
Ballistic, THz	0.05 A/W (uniform field); 0.2 to 0.45 A/W (drop at neck)	none ($A = 1$ at dc)	transit time; where the drop sits

holds in the clean limit only. The ballistic model imposes its potential; the dependence of the terahertz result on the length of the drop is a statement of what a self-consistent calculation would have to decide, and since no drop length gives the shape mechanism an advantage we have not pursued it. The plasmonic model uses a local gate capacitance, which is least accurate for the 175 nm neck under a 30 nm gate; it has ideal contacts except where stated, perfect coupling to an antenna represented by its resistance, and Johnson noise only. It contains no electron heating, and so none of the photothermoelectric response that contributes to measured graphene detectors; the fluid description is quantitative to about 0.3 THz at 100 K and 1 THz at 300 K, where $\omega\tau_{ee} = 1$, and indicative beyond. The two terahertz models describe different idealised devices, a gated fluid at $10^{12}\,\mathrm{cm}^{-2}$ and an ungated ballistic channel at $1.4\times10^{12}\,\mathrm{cm}^{-2}$, compared through their responsivity into the same source; no model of the crossover between them is offered. Finally, the reported Tesla-valve rectification at low current [12] is not reproduced by steady incompressible hydrodynamics and remains unexplained by this work.

VII. CONCLUSION

We have computed the rectification of a graphene funnel and of an electronic Tesla valve in the ballistic, viscous and plasmonic transport limits, from dc to 3 THz and from the small-signal limit to the gate overdrive. The Tesla valve does not rectify ballistically, rectifies in the

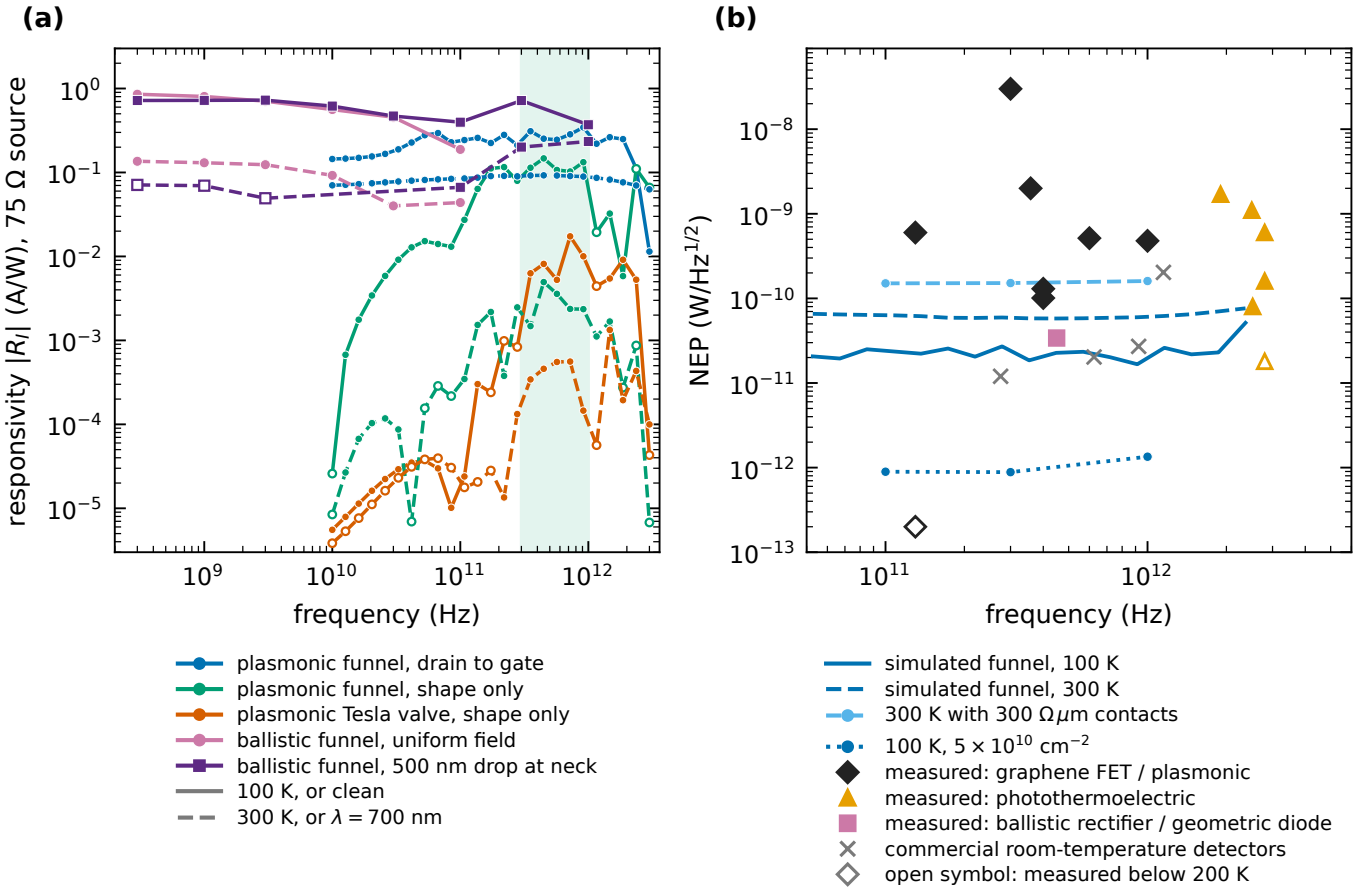


FIG. 8. Synthesis. (a) Current responsivity into a 75 Ω source against frequency for every mechanism: the plasmonic 1 μm funnel with the Dyakonov-Shur termination and with the symmetric drive that isolates the shape, the Tesla valve, and the ballistic funnel with a uniform field and with the potential drop confined to the neck. Solid lines: 100 K parameters or the clean ballistic limit; dashed: 300 K or $\lambda = 700$ nm. (b) Noise-equivalent power of the simulated elements, which is Johnson limited and assumes ideal coupling, against measured graphene terahertz detectors (Table III).

viscous liquid only beyond the drift velocity the Fermi liquid allows, and is an order of magnitude below the funnel at terahertz frequencies. The funnel's terahertz response, 0.07 to 0.34 A/W into a 75 Ω antenna, is the Dyakonov-Shur asymmetry of its terminations; its shape contributes as a matched horn near a taper resonance and not as a diode. The conversion efficiency follows a square law up to a saturation power that scales as the square of the gate overdrive and peaks below one per cent at every density and gate distance; contact resistance attacks the asymmetry itself. The ballistic shape mechanism, driven at 1 THz, survives the transit time only when the potential drop is confined to the neck and then reaches the same responsivity and no more.

Geometry, then, does not make a graphene terahertz rectenna. What it makes is a square-law detector with a Johnson-limited noise-equivalent power of tens of pW/Hz$^{1/2}$ at room temperature and of order 1 pW/Hz$^{1/2}$ at low density at 100 K, if contacts of 100 Ω μm and the assumed scattering times can be had. Two things would change the answer and are outside the models used here: a rectification mechanism that is neither ballistic, viscous nor plasmonic, of which the photothermoelectric effect is the obvious example, and a material whose electron liquid sustains a higher drift velocity relative to its plasma and Fermi velocities.

ACKNOWLEDGMENTS

The simulation code, the analysis and this manuscript were developed with the assistance of an AI coding assistant (Claude, Anthropic); the author directed the work, reviewed the results and takes responsibility for the content. The code, the input geometries, all result files and the scripts that produce the figures are available in the public repository https://github.com/TtheBC01/GeometricDiodes.

TABLE III. Measured graphene terahertz detectors and the simulated element. Responsivity and NEP as stated by the authors; the last column gives the power to which they are referred: the power in a diffraction-limited spot, the full optical beam power, the power delivered to the antenna, an extrinsic figure, the waveguide input, or the available power with ideal coupling. NEPs are estimated from Johnson noise except the one marked with a double dagger, which was measured; the commercial values are typical specifications. Entries marked with a dagger were read from the abstract only.

Reference	Device and mechanism	f (THz)	T (K)	Responsivity	NEP (pW/Hz$^{1/2}$)	Power
[19]	FET, overdamped plasma wave	0.3	300	0.10 to 0.15 V/W	3×10^4	spot
[43]	bilayer FET	0.29 to 0.36	300	1.2 V/W	2×10^3	spot
[44]†	CVD FET, bow-tie antenna	0.6	300	14 V/W	515	optical
[39]	CVD FET, bow-tie antenna	0.4	300	74 V/W	130	optical
[40]	CVD FET, bow-tie antenna	0.4	300	59 to 63 V/W	101 to 114	optical
[45]	encapsulated FET	0.13	300	20 V/W	600	antenna
[20]	encapsulated bilayer FET	0.13	10	>3 kV/W	0.2	antenna
[46]	thermoelectric, patch antenna	1.9	300	4.9 V/W	1.7×10^3 ‡	optical
[41]	photothermoelectric p-n junction	2.5	300	105 V/W	80	spot
[42]	photothermoelectric and plasma wave	2.8	300; 77	49; 180 V/W	160; 18	spot
[7]†	four-terminal ballistic rectifier	up to 0.69	300	764 V/W	34	extrinsic
[49]	commercial zero-bias Schottky	0.22 to 1.1	300	750 to 2200 V/W	5 to 27	waveguide
This work	1 μm funnel, DS, simulated	0.3 to 1	300	0.09 A/W, 52 V/W	60; 150 with contacts	available
This work	same	0.3 to 1	100	0.21 to 0.34 A/W, 36 to 58 V/W	17 to 27	available
This work	same, 5×10^{10} cm^{-2}	0.3	100		0.9	available

Appendix A: Second-order plasmonic response

In units of the contact width W, the plasma speed s_0 and the gated density n_0, Eqs. (4) and (5) contain the friction $\Gamma = W/s_0\tau_{\rm mr}$, the plasma Reynolds number $\mathrm{Re}_s = s_0W/\nu$, the collision time $s_0\tau_{ee}/W$ and the quantum-capacitance fraction q, with the closure $h(n) = 2(1-q)(\sqrt{n}-1)+q\ln n$ and $h''(1) = -(1+q)/2 \equiv \kappa$. Writing $n = 1+\mathrm{Re}(n_1e^{i\Omega t})+n_{20}$ and $\boldsymbol{u} = \mathrm{Re}(\boldsymbol{u}_1e^{i\Omega t})+\boldsymbol{u}_{20}$, the first order is

$$i\Omega n_1 + \nabla\cdot\boldsymbol{u}_1 = 0, \qquad \text{(A1)}$$

$$(i\Omega+\Gamma)\boldsymbol{u}_1 + \nabla n_1 - \nu(\Omega)\nabla^2\boldsymbol{u}_1 = 0, \qquad \text{(A2)}$$

with $\nu(\Omega) = \mathrm{Re}_s^{-1}/(1+i\Omega s_0\tau_{ee}/W)$, and the time average of the second order is

$$\nabla\cdot\boldsymbol{u}_{20} = -\tfrac{1}{2}\mathrm{Re}\,\nabla\cdot(n_1\boldsymbol{u}_1^*), \qquad \text{(A3)}$$

$$\Gamma\boldsymbol{u}_{20} + \nabla P - \mathrm{Re}_s^{-1}\nabla^2\boldsymbol{u}_{20} = -\tfrac{1}{2}\mathrm{Re}\,(\boldsymbol{u}_1^*\cdot\nabla)\boldsymbol{u}_1, \qquad \text{(A4)}$$

where $P = n_{20}+(\kappa/4)|n_1|^2$. The dc contact voltage is $P + c_V|n_1|^2$ with $c_V = 1/8$ for the Dirac closure and zero for shallow water, and the dc particle current through a contact is the flux of $\boldsymbol{u}_{20}+\mathrm{Re}(n_1\boldsymbol{u}_1^*)/2$. Each contact is an equipotential with one unknown voltage per order, closed by Kirchhoff's law for its termination: $RI_c+V_c = V_{a,c}$ for a source of resistance R, $I_c = 0$ for an open contact. The same structure with a matched load at dc gives the rectified power. Voltages are in units of $V_u = m^*s_0^2/e$, resistances in units of m^*s_0/e^2n_0W (65 Ω for $W = 4\,\mu$m), and frequency in units of $s_0/2\pi W$ (0.142 THz).

A contact resistance R_c enters as a change of the ports only: the source becomes a Thevenin source of resistance $R_{\rm ant}+R_c$ and the drain one of zero amplitude and resistance R_c. In second order the dc problem is linear, so the short-circuit current and open-circuit voltage satisfy $I_{\rm sc}(R_{\rm dc}+2R_c) = V_{\rm oc}$ exactly, which is one of the ladder tests. At the signal frequency the antenna does not see the device impedance plus $2R_c$, because the source and drain currents differ by the gate displacement current.

Appendix B: Saturation estimate on the density and gate grid

The peak efficiencies of Fig. 6(b) are $\eta_{\rm peak} = 2r^2(R_{\rm ant}/R_{\rm dc})\,(0.7)^2\,\mathcal{S}$, with r and $R_{\rm dc}$ from the small-signal solver at each grid point and $\mathcal{S}$ the ratio of the time-domain efficiency to the square law at $V_a = 0.7\,V_u$

for the reference device (0.86, 0.89, 0.60 at 100 K and 1.14, 1.10, 0.81 at 300 K for 0.1, 0.3, 1 THz). The assumption that saturation occurs at a fixed V_a/V_u was tested by a time-domain ladder at $(5\times10^{11}\,\mathrm{cm}^{-2}, 10\,\mathrm{nm})$: $\mathcal{S}$ agrees to 0.2% at 300 K and is 0.76 instead of 0.89 at 100 K, so the 100 K peak values at low density and thin gates are upper estimates by 10 to 20%.

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